A Deconvolutional Technique Based on Pseudoinverse for Online Energy Reconstruction

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***Abstract*—Modern High-Energy Physics (HEP) experiments face significant challenges in data processing due to high-luminosity environments where the overlapping of signals, known as pile-up, degrades detector resolution. Traditionally, the Optimal Filter (OF) algorithm is employed for amplitude estimation. However, its performance is limited in high-occupancy scenarios because the filter weights must be specifically recalculated for different pile-up levels to maintain accuracy. This paper proposes an alternative online energy reconstruction technique based on deconvolutional filtering using the Moore-Penrose pseudoinverse. By treating the signal reconstruction as an inverse problem, the proposed method relies solely on the knowledge of the system's impulse response to effectively separate overlapping pulses across varying pile-up conditions. Unlike the OF approach, which typically requires an additional peak detection stage combined with a phase filter, the deconvolution output directly provides the energy values at the precise time of the collision. The approach was validated using the Lorenzetti Showers framework, simulating Liquid Argon (LArg) calorimeter responses under severe multi-hit conditions. Results demonstrate that the deconvolution method significantly reduces the Root Mean Square Error (RMSE) from approximately 200 MeV in the conventional OF to values between 50 and 100 MeV, representing a reduction of up to 62.5% in reconstruction error. Also, it maintains high detection efficiency compared to the conventional OF. The proposed architecture is suitable for low-latency implementation in Field Programmable Gate Arrays (FPGAs), offering a streamlined and robust solution for real-time trigger systems.**

***Index Terms*—Deconvolution, Calorimetry, High-energy physics (HEP), Online energy reconstruction, Pile-up**

I. Introduction

High-energy physics (HEP) experiments rely fundamentally on advanced instrumentation to characterize known processes and search for new physics. In particle colliders, calorimeters play a primary role by measuring the energy and position of incoming particles through fine-grained segmentation. Because these collisions generate an immense volume of data, experiments employ multi-tiered trigger systems to filter out background noise and select relevant physics events [1]–[3].

Calorimeter acquisition systems are essential at all trigger levels due to their fast response times. Typically, a conditioning circuit (shaper) provides a fixed-shape readout signal proportional to the amplitude. In the fast, hardware-based first-level trigger, the system must continuously perform two critical free-running tasks: detecting the signal time-stamp and estimating its amplitude. Traditionally, this is achieved using linear FIR (Finite Impulse Response) filters optimized via techniques such as OF [4], [5], followed by peak detection circuitry.

As modern and future colliders, such as the High-Luminosity LHC [1] and the Future Circular Collider (FCC) [6], push for unprecedented luminosity levels, detectors face increasingly harsh high-occupancy environments [2]. The primary challenge in this scenario is *pile-up*, the superposition of signals that occurs when the time interval between consecutive interactions is shorter than the calorimeter's time response [7].

Severe signal overlap significantly degrades the shape of the original pulse, compromising standard linear estimation methods. OF techniques require the underlying pulse shape to remain intact to achieve accurate amplitude and phase measurements. When pile-up occurs, this shape is lost. While current baseline designs adapt OF techniques to mitigate pile-up noise, these methods are still highly dependent on precise pulse shape [5]. This necessitates a more robust approach that can be resilient to severe pile-up conditions.

To overcome the limitations regarding pile-up in conventional methods, this work proposes a novel free-running signal deconvolution technique based on the Moore-Penrose pseudoinverse. By treating the pulse reconstruction as an inverse problem, the proposed method shifts away from the local constraints of traditional OF.

Instead of estimating a single local amplitude, this framework computes a deconvolution matrix that approximates the inverse of the detector's convolution matrix, allowing for the simultaneous estimation of a stream of energy deposits. Furthermore, by applying a predefined threshold to eliminate non-physical stochastic fluctuations, the method effectively concentrates the signal into sharp pulses, achieving a Sparse Representation of the collision energy [7]. This architecture offers two critical advantages for real-time trigger systems:

- **Impulsive Signal Recovery:** The deconvolution process inherently transforms the overlapped waveforms back into sharp, impulsive signals. This allows for direct and precise localization of the particle's arrival time without requiring an extra peak detection circuit. In contrast, OF methods necessitate a secondary peak detector circuitry to identify the maximum value and complement the filtering stage.
- **Pile-Up Independent Design:** The deconvolution operator relies exclusively on the knowledge of the reference pulse shape. Once the inverse model is established, it performs effectively under any pile-up scenario. Conversely, standard OF techniques require the recalculation of specific weight coefficients for every distinct pile-up level in order to properly model the varying noise covariance.

To thoroughly evaluate and validate the proposed pseudoinverse method under realistic conditions, this work relies on datasets generated by the *Lorenzetti Showers* framework [8]. By accurately simulating Liquid Argon (LArg)-like bipolar pulses, severe multi-hit pile-up, and electronic noise, the framework ensures that the deconvolution algorithm is tested under conditions that match the demanding environments of modern high-throughput calorimeters.

The remainder of this paper is organized as follows. Section II reviews the conventional Optimal Filtering (OF) method and discusses its main limitations when operating under severe high-occupancy and pile-up conditions. Section III introduces the proposed deconvolutional technique based on the Moore-Penrose pseudoinverse, highlighting its advantages for impulsive signal recovery and pile-up independent design. Section IV details the experimental setup, describing the data generation and the LArg-like pulse modeling using the Lorenzetti Showers framework. Section V presents the performance evaluation, comparing the proposed method with the standard OF in terms of detection capability and energy estimation accuracy. Finally, Section VI draws the main conclusions of this work and outlines future perspectives for hardware implementation.

II. Conventional Energy Estimation: The Optimal Filtering Method

The precise reconstruction of energy and timing information from digitized pulses is a fundamental requirement in modern calorimeter instrumentation. Currently, the most widely adopted technique for this purpose is the Optimal Filter (OF) algorithm [9]. The OF is a Digital Signal Processing (DSP) method specifically designed to estimate the amplitude (A) and the arrival time (τ) of an analog collision signal measured from a set of m discrete samples ($y_i,\ i = 1, \ldots, m$), obtained at fixed time intervals (t_i), forming the vector $\mathbf{y} = [y_1, y_2, \ldots, y_i, \ldots, y_m]^T$. By employing a weighted linear combination of these samples, the algorithm aims to maximize the signal-to-noise ratio, effectively minimizing the degradation of energy resolution caused by both intrinsic electronic noise and the overlapping of pulses (the pile-up). Consequently, the OF serves as a benchmark for real-time signal processing in high-throughput detector environments.

A normalized pulse shape function is modeled by the detector's impulse response, denoted as $h(t)$. This continuous function can be sampled at the same time instances t_i, yielding the discrete samples $h(t_i) = h_i$, which compose the reference vector $\mathbf{h} = [h_1, h_2, \ldots, h_i, \ldots, h_m]^T$. Consequently, the observed digitized signal samples y_i can be expressed as:

$$y_i = Ah(t_i - \tau) + n_i, \tag{1}$$

where A is the pulse amplitude, representing the deposited energy, τ represents the time shift (phase) relative to the sampling clock, and n_i accounts for the noise samples, forming vector $\mathbf{n} = [n_1, n_2, \ldots, n_i, \ldots, n_m]$. Since the instrumentation operates at high rates (e.g. 40 MHz), the OF assumes small time deviations, allowing the use of a first-order Taylor expansion:

$$y_i \simeq Ah(t_i) - A\tau h'(t_i) + n_i. \tag{2}$$

Based on this approximation, the amplitude $A = \langle u \rangle$ and the time-weighted amplitude $A\tau = \langle v \rangle$, where $\langle \cdot \rangle$ stands for the expected value, are reconstructed through a linear combination of the samples y_i using specific weights a_i and b_i:

$$u = \sum_{i=1}^{m} a_i y_i \quad \text{and} \quad v = \sum_{i=1}^{m} b_i y_i. \tag{3}$$

The coefficients a_i and b_i are calculated to minimize the variance of the estimator while satisfying the constraints $\sum a_i h_i = 1$, $\sum a_i h'_i = 0$, $\sum b_i h_i = 0$, and $\sum b_i h'_i = -1$. This ensures that, in the absence of noise, u yields the exact amplitude A. A detailed derivation of these weights and the noise covariance matrix handling can be found in [9].

Despite its efficiency in stochastic noise reduction, the performance of the OF is significantly limited in high-luminosity scenarios. The conventional algorithm treats pile-up as an additional noise component, attempting to mitigate its effects by incorporating its statistical properties into the variance minimization process. However, since the OF is a second-order estimator, it only accounts for the covariance structure of the interference. In high-occupancy environments, pile-up signals exhibit non-Gaussian characteristics and higher-order statistical information that remain inaccessible to the OF formalism. Consequently, the method fails to effectively suppress the non-stationary distortions caused by overlapping pulses, leading to significant systematic errors in energy estimation. These limitations necessitate a more robust approach, such as deconvolutional filtering, which can resolve individual pulses without relying on the underlying noise statistics.

III. Proposed Approach: Sparse Representation via Pseudoinverse Deconvolution

To overcome the limitations regarding the conventional methods, this work proposes a signal reconstruction technique based on deconvolutional filtering. By treating the pulse reconstruction as an inverse problem, the system operates with higher resiliency to the pile-up level.

A. Mathematical Framework

The digitized signal samples y_i, which compose the observed vector $\mathbf{y}$, can be modeled as the convolution between the detector's impulse response $\mathbf{h}$ and the sequence of energy depositions $\mathbf{a}$. While the conventional OF approach treats the energy as a local scalar amplitude A, the proposed method considers an amplitude vector $\mathbf{a} = [A_1, A_2, \dots, A_i, \dots]$ to enable continuous, free-running reconstruction. In this framework, $\mathbf{a}$ represents a stream of energy deposits over time.

Under this discrete-time representation, the signal model is expressed in matrix form as:

$$\mathbf{y} = \mathbf{H}\mathbf{a} + \mathbf{n}, \tag{4}$$

where $\mathbf{H}$ is the convolution matrix constructed from time-shifted versions of the reference pulse shape $\mathbf{h}$:

$$\mathbf{H} = \begin{bmatrix} h_1 & 0 & 0 & 0 & \cdots & 0 \\ h_2 & h_1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ h_m & \cdots & h_i & \cdots & h_2 & h_1 \\ 0 & h_m & \cdots & h_i & \cdots & h_2 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & h_m \end{bmatrix}, \tag{5}$$

and $\mathbf{n}$ represents the electronic noise (Gaussian white noise).

To reconstruct the energy amplitudes $\mathbf{a}$ from the measured samples y_i, we seek an operator that reverses the effect of the system's response. Since the system is typically overdetermined, a direct inverse $\mathbf{H}^{-1}$ does not exist. Instead, the optimal solution in the least-squares sense is obtained via the Moore-Penrose pseudoinverse [10]. Thus, the reconstructed amplitude vector $\hat{\mathbf{a}}$ can be calculated as:

$$\hat{\mathbf{a}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{y} = \mathbf{W}\mathbf{y}, \tag{6}$$

where $\mathbf{W} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T$ is defined as the deconvolution matrix that approximates the inverse of $\mathbf{H}$. In a theoretical framework, the entire matrix $\mathbf{W}$ could be used to reconstruct the energy vector. However, for real-time instrumentation in a free-running mode, it is more efficient to implement the deconvolution as a single Finite Impulse Response (FIR) filter.

Each row of the matrix $\mathbf{W}$ can be interpreted as a set of deconvolution coefficients with a specific reconstruction delay. In this work, rather than employing the full matrix, a single row is selected to serve as the FIR filter kernel. The choice of the optimal row is determined empirically, aiming to minimize the estimation error while maintaining the lowest possible latency (delay). Typically, rows near the center of the matrix provide the best balance between noise suppression and temporal precision. By applying this single-row filter continuously to the incoming data stream, the system achieves high-throughput energy reconstruction with fixed latency.

By applying this optimized deconvolutional filter to the observed signal $\mathbf{y}$, the noise term $\mathbf{n}$ is processed alongside the physical signal. Since it's a Gaussian noise, the Moore-Penrose pseudoinverse provides the Best Linear Unbiased Estimator (BLUE) for this overdetermined system, minimizing the sum of squared residuals. While the impact of electronic noise on the energy estimation $\hat{\mathbf{a}}$ is statistically minimized, it is not entirely eliminated. Instead, stochastic fluctuations may appear as low-amplitude oscillations in the reconstructed vector $\hat{\mathbf{a}}$, even in time instances where no physical energy deposition occurred. Nevertheless, by applying a predefined threshold, below which the reconstructed energy is considered to have no physical interest, these fluctuations can be effectively eliminated, recovering the sparse characteristic of the original data.

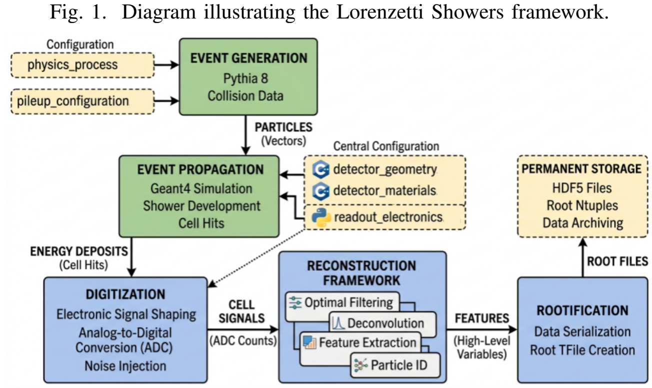
IV. Experimental Setup: The Lorenzetti Framework

To evaluate the proposed pseudoinverse deconvolution approach, simulated datasets were generated using the *Lorenzetti Showers* framework [8]. Lorenzetti is a general-purpose, integrated software framework specifically designed to support the development of novel signal reconstruction and triggering strategies for segmented calorimeters. Unlike collaboration-restricted software, it provides comprehensive low-level access to calorimeter cell readouts, incorporating precise models of the entire electronic signal-processing chain.

The data generation pipeline in Lorenzetti is highly modular. It primarily utilizes Pythia 8 to generate particle collisions, producing both the main physics processes and background minimum-bias events, and Geant4 to propagate these particles through the modeled detector geometry. This propagation computes the exact true energy deposited (hits) within each calorimeter cell for a given bunch crossing (BC). In the Figure 1, you can see the diagram of the Lorenzetti Showers framework.

To accurately replicate the high-occupancy environments of modern colliders, Lorenzetti incorporates a dedicated pile-up merging module. This module superimposes multiple minimum-bias events onto the main signal based on a configurable average number of interactions ($\langle\mu\rangle$). This procedure effectively simulates both in-time and out-of-time pile-up effects, spanning the energy deposits across several consecutive bunch crossings and creating the continuous, heavily overlapped signal streams required to test the free-running robustness of the DSP algorithms.

Fig. 1. Diagram illustrating the Lorenzetti Showers framework.



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For evaluating signal processing techniques, the framework's digitization stage is of primary importance. This module emulates the front-end analog-to-digital conversion (ADC) by transforming discrete energy hits into continuous-time electronic signals.

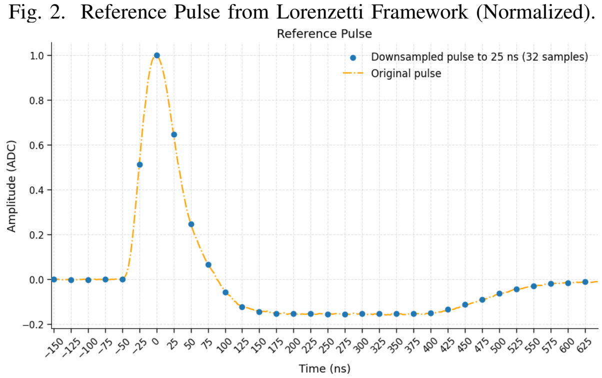
To model the characteristic response of Liquid Argon (LArg) calorimeters widely used in HEP experiments, the framework was configured to apply a bipolar pulse-shaping function to the energy deposits. This pulse shape mimics the standard conditioning circuits (shapers) that optimize the signal-to-noise ratio and standardize the pulse width and amplitude before digitization. Furthermore, the digitization module integrates realistic experimental constraints, including stochastic electronic noise and random time shifts (phase misalignments), ensuring that the generated signal stream accurately represents the demanding conditions faced by real-time online trigger systems.

V. Performance Evaluation and Results

The performance of the proposed pseudoinverse-based deconvolution method was evaluated using simulated calorimeter signals generated with the Lorenzetti Framework. This section presents both qualitative and quantitative analyses, focusing on signal reconstruction under different pile-up conditions, detection capability, and energy estimation accuracy.

The reference pulse shape used in this study is derived from the Lorenzetti digitization stage, which emulates the bipolar response of the calorimeter. The continuous signal is downsampled to a 25 ns sampling period, resulting in a discrete representation with 32 samples, as shown in Figure 2.

Fig. 2. Reference Pulse from Lorenzetti Framework (Normalized).



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To illustrate the impact of pile-up, Figure 3 shows the superposition of two pulses with different amplitudes and temporal offsets. This scenario represents a simplified multi-hit condition where two consecutive energy deposits overlap within the detector response window.

The corresponding OF processing chain for this example of signal superposition is shown in Figure 4. The first FIR filter (OF A) computes the reconstructed amplitude A through a weighted sum of the input samples, providing an estimative that is valid only when the peak of the signal is in filter's central tap. In parallel, the second FIR filter (OF B) evaluates

Fig. 3. Signal superposition.

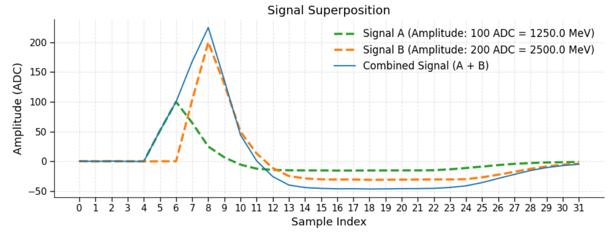

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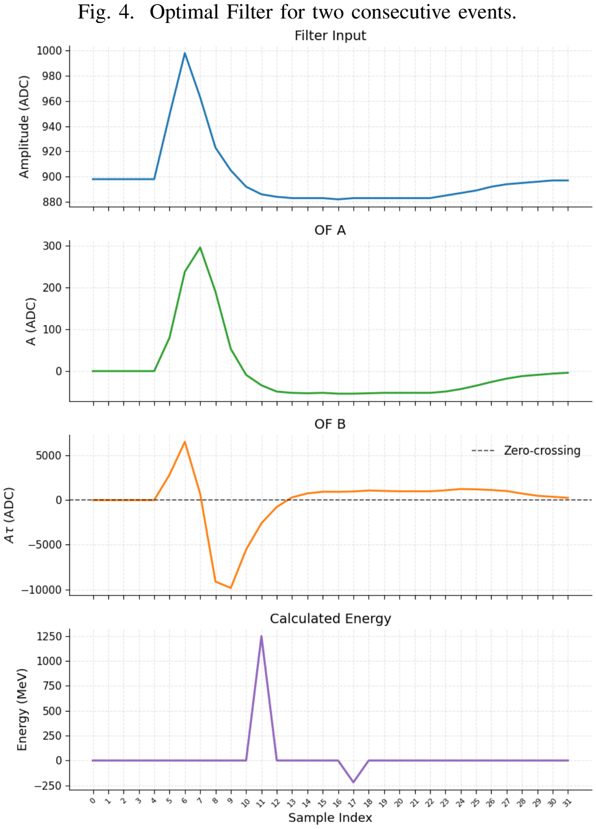
Fig. 4. Optimal Filter for two consecutive events.



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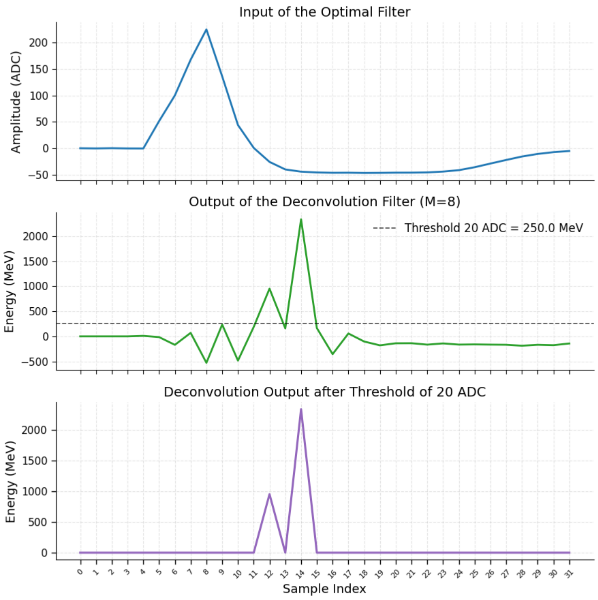
$A\tau$, which encodes the temporal deviation of the pulse relative to the reference shape. This way it's possible to extract the timing information, where the pulse peak is identified through the zero-crossing condition of the filter OF B.

In this example, after the main pulse is detected, the undershoot of the bipolar response produced an additional zero-crossing, leading to a secondary detection at sample index 19. The reconstructed output also demonstrates that only a single pulse is identified despite the presence of two distinct energy depositions. This behavior reflects an intrinsic characteristic of the OF approach, which is optimized for isolated pulse reconstruction. As a result, closely spaced signals are more challenging to resolve.

In contrast, the deconvolution approach for the same input signal, shown in Figure 5, produces two distinct peaks aligned with the original energy deposits. The result shown is for a FIR filter designed with 8 coefficients. It demonstrates the ability of the proposed method to separate overlapping signals by effectively inverting the convolution process and recovering a sparse representation of the input. In this case, the choice of the threshold (250 MeV) was arbitrary.

Fig. 5. Deconvolution for two consecutive events.



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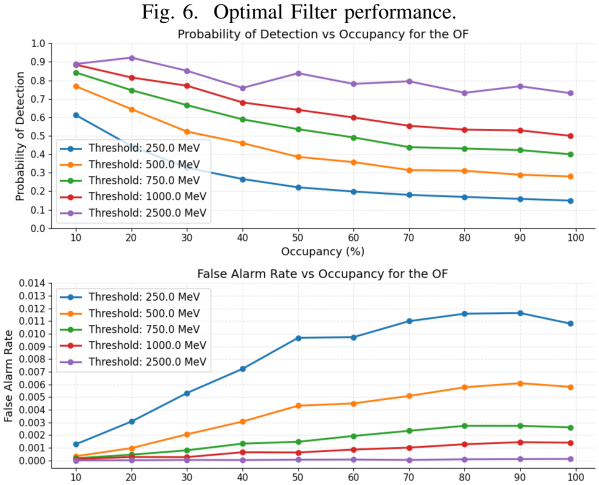
The quantification of the detection capability of both methods is evaluated using the Probability of Detection (PD) and the Probability of False Alarm (PFA). The PD is defined as the fraction of true energy depositions that are correctly identified, while the PFA corresponds to the fraction of zero-signal samples that are incorrectly classified as events.

The detection performance of the OF under different levels of pile-up and threshold is presented in Figure 6. As occupancy increases (at pile-up level), a degradation for the detection is observed, accompanied by an increase in false alarms due to signal distortion.

Fig. 6. Optimal Filter performance.



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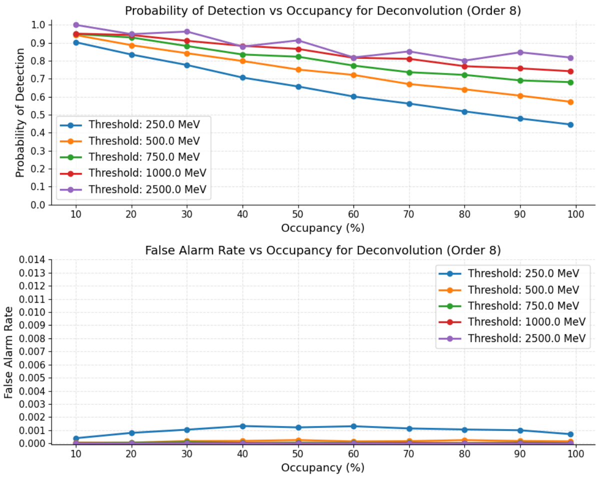
The performance of the deconvolution method for different filter orders compared is shown in Figure 7. The results indicate that higher-order filters improve detection efficiency (higher PD) and the deconvolution approach maintains robust detection performance even under high pile-up conditions.

Fig. 7. Deconvolution performance.



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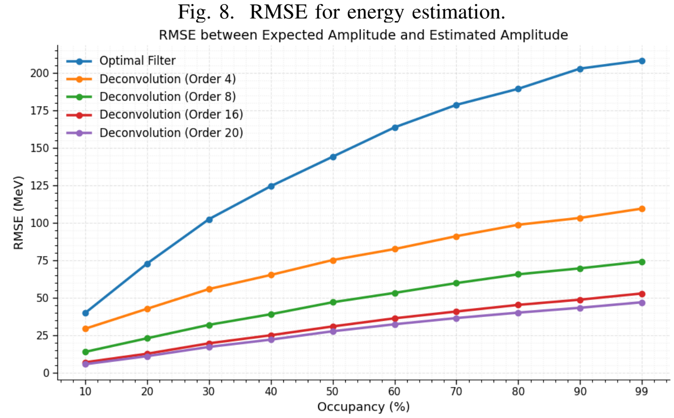
The accuracy of energy estimation is evaluated using the Root Mean Square Error (RMSE) and the mean error of the energy reconstruction. Figure 8 shows the RMSE as a function of operating conditions. The OF exhibits an increased reconstruction error as the pile-up intensifies, reflecting its sensitivity to overlapping pulses and phase misalignment. In contrast, the deconvolution method achieves lower RMSE values, particularly for higher filter orders, due to its ability to decouple overlapping contributions and recover individual energy deposits.

Fig. 8. RMSE for energy estimation.



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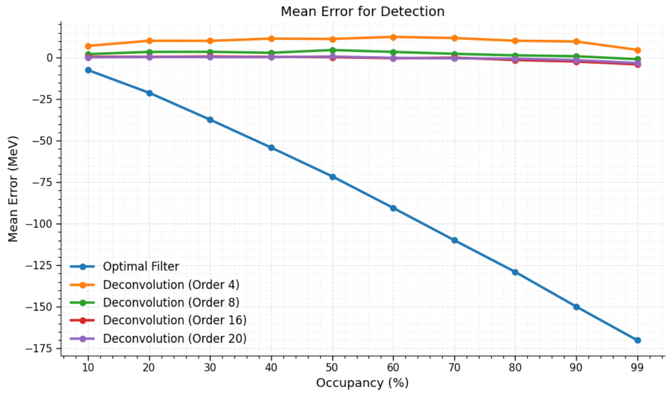
The mean error for the energy reconstruction is presented in Figure 9. The OF shows a systematic bias, especially in high-occupancy scenarios, where overlapping pulses lead to underestimation or misallocation of energy.

Fig. 9. Error mean for OF and Deconvolution.



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VI. Conclusion

This paper presented a novel approach for online energy reconstruction in high-throughput calorimeters using a deconvolutional filter based on the Moore-Penrose pseudoinverse. The proposed method shifts the reconstruction paradigm from local template matching to a continuous-flow inverse problem, specifically designed to operate in high-occupancy environments.

The experimental results obtained through the Lorenzetti framework demonstrate that the deconvolutional technique consistently outperforms the conventional Optimal Filter (OF) across all evaluated occupancy levels. In terms of detection capability, the Probability of Detection (PD) for the deconvolutional method reached 0.80 at 10% occupancy, significantly higher than the 0.35 achieved by the OF. Even under extreme conditions, such as 99% occupancy, the proposed method maintained a PD above 0.60, while the OF performance dropped to 0.14. Regarding Probability of False Alarm (PFA), the deconvolution method proved more robust, maintaining a PFA of 0.02 at 10% occupancy and remaining below 0.10 at 99% occupancy, whereas the OF showed higher error rates of 0.05 and 0.19, respectively. Furthermore, the accuracy of energy estimation saw a substantial improvement, with the Root Mean Square Error (RMSE) being reduced from approximately 200 MeV in the OF to a range between 50 and 100 MeV using the proposed filter.

Beyond performance metrics, the deconvolutional approach offers a more streamlined and robust instrumentation design. Unlike the OF, which requires the continuous recalculation of filter weights based on varying pile-up noise statistics, the deconvolutional filter relies solely on the knowledge of the system's impulse response $h(t)$. Additionally, the proposed architecture simplifies the processing chain by replacing the complex zero-crossing detection circuits required by the OF with a single FIR structure followed by a simple thresholding stage. Despite these advantages, this work is limited by the assumption of a static pulse shape and the evaluation of the algorithm in a CPU-based simulation environment, without considering the effects of clock jitter or signal phase variations.

Future research will focus on the implementation of the deconvolution kernel in Field Programmable Gate Arrays (FPGAs) to assess hardware resource utilization and latency. Additionally, further studies will investigate adaptive thresholding techniques and the robustness of the method against pulse shape distortions in next-generation calorimetry systems.

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